Safety risk evaluation based autonomous vehicle decision-making approach for cut-in emergency scenario

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***Abstract*—Improving the risk avoidance abilities of autonomous vehicles in corner scenarios is crucial to ensure driving safety. Emergency cut-in traffic scenarios are typical and dangerous, and it is challenging for autonomous vehicles to complete behavior decision-making in this scenario. In this study, we present a decision-making strategy for autonomous vehicles that evaluates safety risks specifically for emergency cut-in scenarios. The approach adopts an Oriented Bounding Box (OBB) collision detection method based on the Separate Axis Theorem (SAT). Additionally, it considers the transient process of the cut-in scenario to predict the behavior of the cut-in vehicle and complete the collision threat assessment. The approach also designs a decision-making state machine model to maintain a safe distance with the cut-in vehicle. Furthermore, the designed decision-making approach is validated using Prescan, Matlab/Simulink, and Carsim joint simulation based on real traffic accident data collected by NHTSA, combined with planning and control algorithms. The simulation results demonstrate that the autonomous vehicle decision-making approach designed in this paper is feasible in emergency cut-in traffic scenarios and exhibits a better risk avoidance effect than the state machine model that only considers adaptive following.**

***Index Terms*—Autonomous Vehicle, Cut-In Emergency Scenario, Decision-Making Approach, Safety Risk Evaluation**

I. Introduction

The advancement of autonomous driving technology plays a critical role in reducing traffic accidents, with the decision-making system serving as the ”driving brain” that guides the vehicle’s actions. The reasonableness of its decision-making will be directly related to the safety and comfort of autonomous vehicles, but also related to the degree of intelligence of autonomous vehicles. In all traffic accidents, cut-in traffic scenarios are typical and dangerous, which are prone to rear-end collision, oblique angle collision, and other accidents, and the incidence rate of this condition accounts for 8% of total accidents [1]. So, it is necessary to evaluate the relative distance between the ego vehicle and the cut-in vehicle in real-time, and to make a reasonable and timely maneuver to ensure the safety of driving. But the current coping strategies are relatively brief and single. Therefore, it is crucial to study the behavioral decision-making approach for autonomous vehicles in cut-in scenarios.

Numerous studies have focused on assessing driving safety risks, often utilizing traffic conflict indicators as surrogate safety measures to evaluate the severity of each conflict scenario [2]. The most commonly used indicators include measures of spatial and temporal proximity, such as Time-to-Collision (TTC) and Time-Headway (THW). In addition to these, other metrics that reflect the vehicle’s own motion characteristics, such as Deceleration Rate to Avoid a Crash (DRAC), are also considered as they quantify the vehicle’s capacity to prevent collisions. Meanwhile, significant research efforts are directed toward developing decision-making algorithms for autonomous driving, with widely adopted models including the rule-based behavioral decision-making model, exemplified by finite-state machines, and the learning-based behavioral decision-making model, represented by reinforcement learning algorithms [3]. Rule-based behavioral decision-making models are simple and have good real-time performance, but they struggle to determine state boundaries in complex traffic conditions and have limitations in improving their logical structure. Learning algorithm-based models have rapidly developed in recent years, performing well in specific conditions and tasks, but they rely heavily on extensive training data and face challenges in multi-scenario and multi-task coordination. The finite-state machine belongs to the rule-based behavioral decision-making algorithms, usually setting basic driving behaviors as a finite set of states and taking the vehicle’s driving state as the triggering condition between driving behavior patterns. Autonomous vehicles complete state transitions by receiving triggering information, thereby making behavioral decisions [4]. Despite these advancements, decision-making for autonomous vehicles in cut-in scenarios remains challenging. The transient process of the cut-in scenario must be considered, and the cut-in behavior must be predicted to dynamically adjust the decision-making behavior of autonomous vehicles. This ensures that they can maintain

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a safe and smooth distance from the cut-in vehicle and effectively respond to various emergency cut-in behaviors.

In this paper, a behavioral decision-making strategy for autonomous vehicles considering emergency cut-in traffic scenarios is proposed, combining planning and control algorithms, based on real traffic accident data collected by NHTSA, and the designed decision-making approach is simulated and verified by using Prescan&Matlab/Simulink&Carsim simulation software. Due to the inability of traditional traffic conflict indicators to detect collision risks during lane-changing, we have implemented an Oriented Bounding Box (OBB) collision detection method based on the Separation Axis Theorem (SAT). This decision-making approach takes into account the transient process of cut-in scenarios and predicts the cut-in behavior. To achieve behavioral decision-making, a decision-state machine model is built. This enables the autonomous vehicles to dynamically adjust the relative distance between themselves and cut-in vehicles. Consequently, autonomous vehicles can successfully avoid safety risks in response to emergency cut-in traffic scenarios. Compared to state machine models that only consider adaptive vehicle following, this approach exhibits a better risk avoidance effect.

The main contributions of this paper are summarized as follows: (1) Utilizing the Oriented Bounding Box (OBB) collision detection method to accurately estimate the collision probability during the cut-in process, which serves as a conditional constraint for the design of the decision-state machine model; (2) Accounting for the transient nature of the cut-in scenario and predicting cut-in behavior to employ various temporal indicators for comprehensive collision risk assessment, enabling safe and smooth execution of the risk avoidance strategy; (3) Constructing a rear-end collision scenario in simulation software based on real-world traffic accident data to validate the feasibility and effectiveness of the proposed decision-making approach.

In the rest of the paper, Section II analyses the OBB collision detection method and the longitudinal prediction distance, and builds a decision-state machine model. Section III briefly describes the LQR-based and dual PID control methods used, and Section IV gives the data of the real traffic accident scenarios and the experimental scenarios and result diagrams of the joint simulation in Prescan&Matlab/Simulink&Carsim. joint simulation of the experimental scenarios and schematic results. Section V provides conclusions.

II. Finite State Machine Based Decision-making Model

To achieve effective decision-making for autonomous vehicle safety in emergency cut-in scenarios, this section first introduces various traffic conflict and safety assessment indicators, with a particular focus on the OBB collision detection method and the longitudinal prediction distance developed in this study. These indicators are then employed as state transition conditions to design a state machine-based decision-making strategy. The architecture of the decision-making module is illustrated in Fig. 1.

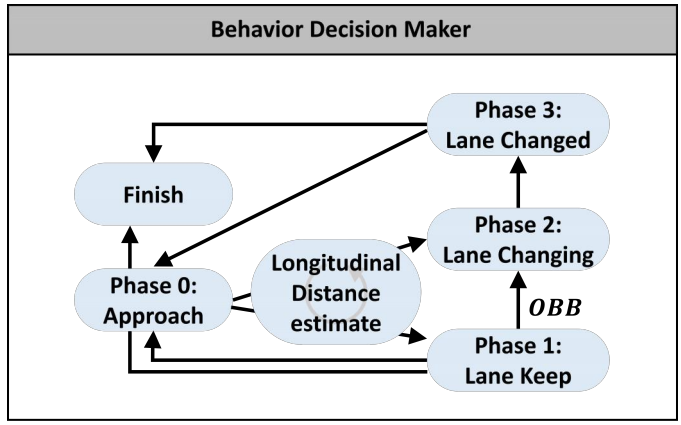

Fig. 1. Decision algorithm showing the state-transition diagram for a cut-in process

A. State transfer indicator system

The state transition indicators used in constructing the decision-state machine algorithm include traffic conflict time indicators (such as TTC and THW), the speed difference between the ego vehicle and the cut-in vehicle (ΔV),, the safety risk index (SI), and the longitudinal prediction distance (d). Among these, the decision-state machine determines which time indicators to utilize for evaluating collision risk based on the longitudinal prediction distance, denoted as 'd'. This section provides a brief overview of the role of 'd', with detailed explanations to be given in the third subsection.

Time to Collision (TTC) is defined as "the time remaining before a collision occurs between two vehicles, assuming both maintain their current course and speed difference [5]." Generally, a smaller TTC value indicates a higher risk of collision. The formula for calculating TTC is as follows:

$$TTC = \frac{D}{V_{rel}} \tag{1}$$

Where D represents the relative distance between the two vehicles, and V_{rel} denotes their relative speed.

Time Headway (THW) is another indicator used to evaluate the criticality of a traffic scenario. It measures the time difference between the front and rear vehicles passing through the same location. THW is typically calculated by dividing the distance between the front and rear vehicles by the speed of the rear vehicle [6]. The calculation formula is as follows:

$$THW = \frac{D}{V} \tag{2}$$

Where D is the relative distance between the two vehicles and V is the speed of the rear vehicle.

The difference between the desired speed of the ego vehicle and the speed of the vehicle in front ΔV, is:

$$\Delta V = V_d - V_c \tag{3}$$

Where the ego vehicle desired velocity is V_d, and the cut-in vehicle velocity is V_c.

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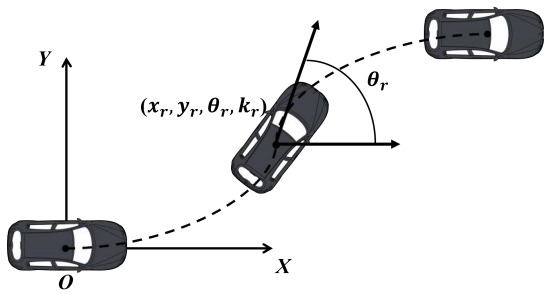

Fig. 2. Autonomous vehicle trajectory planning

In addition, considering that the trend of TTC is opposite to the collision risk and vehicle safety status, this paper introduces a safety risk index, referred to as Safety Risk Index (SI), to evaluate the magnitude of risk in the event of a longitudinal collision between vehicles. This index is calculated using the following formula:

$$SI = \frac{V_{\text{rel}}}{D} \tag{4}$$

B. Oriented Bracketing Box (OBB) collision detection

In the emergency cut-in traffic scenarios, because the driving directions of the cut-in vehicles and the ego vehicles are not parallel, it is impossible to determine whether there is a risk of collision by calculating TTC or THW. Therefore, it is necessary to use convex polygonal collision detection techniques to predict whether a collision will occur between these two vehicles. In this paper, we choose the OBB collision detection algorithm based on the SAT, which constructs a directed enveloping box based on the center and size of the object as well as the axis of rotation, which fits the rotating object better than the Axis-aligned Bounding Box (AABB) enveloping box [7].

This part of the study starts with trajectory planning for two vehicles, in which the trajectory planning for the ego vehicle is simplified, i.e., assuming that the vehicle is driving straight ahead at a constant speed in the direction of the current speed or changing lanes at a certain heading angle, and using a fifth-degree polynomial for the cut-in vehicle. The system establishes the geodetic coordinate system XOY as shown in Fig. 2, and adopts the positional coordinates of the center of gravity of the vehicle (x, y) to describe the vehicle's trajectory [8].

Two functions of time t can be obtained using a fifth-degree polynomial planning trajectory, calculated as follows,

$$x(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3 + a_4 t^4 + a_5 t^5 \tag{5}$$

$$y(x) = b_0 + b_1 x + b_2 x^2 + b_3 x^3 + b_4 x^4 + b_5 x^5 \tag{6}$$

Where $a_0, a_1, a_2, \ldots, a_5$, $b_0, b_1, b_2, \ldots, b_5$ are the coefficients to be determined. The starting and ending information of the driving trajectory of the autonomous driving vehicle is obtained through calculation, from which the corresponding coefficients in Eqs. (11) and (12) can be solved. On this basis,

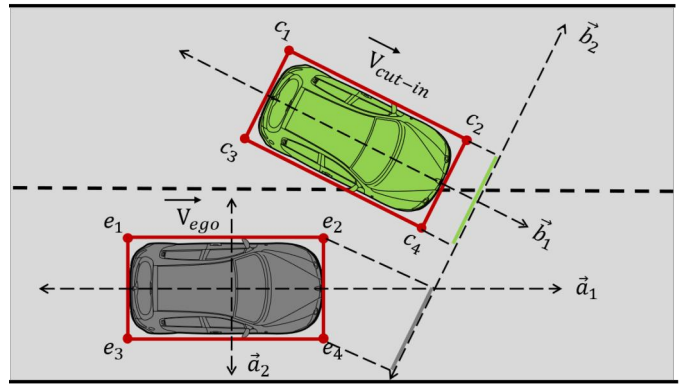

Fig. 3. Schematic of collision detection during the cut-in process

y(x) is converted into y(t) to obtain the expression of the horizontal and vertical coordinates of the desired trajectory concerning the time t. At the same time, by derivation, the trajectory tangent angle θ_r and curvature k_r can be derived from it, and the computational formula is as follows:

$$\theta_r(t) = \arctan\{y_r'[x_r(t)]\} \tag{7}$$

$$k(t) = y_r''[x_r(t)]\left\{1 + [y_r'[x_r(t)]]^2\right\}^{-\frac{2}{3}} \tag{8}$$

Then the collision detection is completed using OBB. Firstly, the box is constructed based on the vehicle and the main axes of the two boxes are found, i.e., the two sides (i.e., length direction and width direction) of the two rectangles, which are set as $\vec{a_1}$, $\vec{a_2}$, $\vec{b_1}$, $\vec{b_2}$. Next, the projections are calculated, i.e., the projection of each OBB on each axis can be calculated by the dot product of points to axes [9]. Assuming that the four vertices of the ego vehicle OBB are e_1,e_2,e_3,e_4, and the four vertices of the cut-in vehicle OBB are c_1,c_2,c_3,c_4, the projected points of the ego vehicle on axis $\vec{a}$ is:

$$\text{proj}_{\vec{a}}(\vec{v}) = \frac{\vec{e} \cdot \vec{a}}{\|\vec{a}\|} \tag{9}$$

A specific representation is shown in Fig. 3.

The projections of the ego vehicle and the cut-in vehicle on the four coordinate axes are calculated separately, after which the minimum and maximum values of the projection points of the bounding box on each axis are found to obtain the boundary or range of the projections on the 2D plane. Afterwards, the projection overlap is checked for each time step δ_t (assumed to be 0.1s) during the cut-in process. If the projections on all axes overlap at the end of a certain time step, the two vehicles are determined to collide, otherwise no collision occurs. Take this figure as an example, at this moment in time the projections of the two vehicles on the $\vec{b_2}$ axes do not overlap and do not satisfy all projections overlap, i.e., the vehicles do not collide.

C. Longitudinal prediction distance d

Given the high-risk nature of front-vehicle emergency cut-in scenarios, it is essential to consider the transient characteristics

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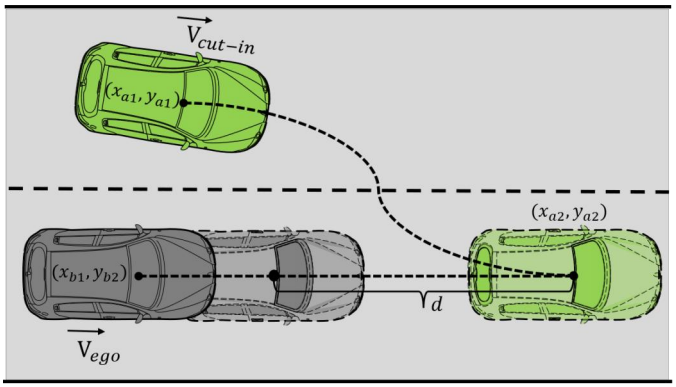

Fig. 4. Schematic of longitudinal distance prediction during the cut-in process

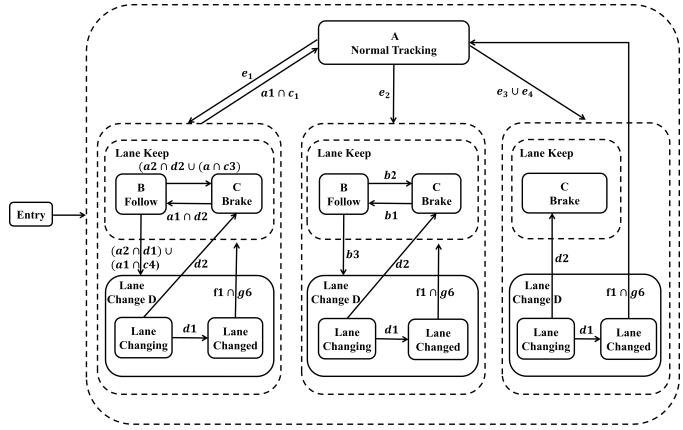

Fig. 5. State machine decision mode switching diagram

of such situations, predict the cut-in vehicle's behavior, and dynamically adjust the autonomous vehicle's decision-making to maintain a safe distance from the cut-in vehicle in a smooth and secure manner, effectively handling different types of emergency cut-ins.

Building upon the trajectory planning analysis from the previous subsection, when the lateral and longitudinal velocities, accelerations, starting positions of both the ego vehicle and the cut-in vehicle, as well as the cut-in vehicle's longitudinal position, are known, the longitudinal distance (d) between the two vehicles after the cut-in process can be estimated. This estimation helps determine the appropriate traffic conflict indicator to be used as a constraint for state transitions and behavioral decision-making, as shown in Fig. 4.

The specific cut-in behavior prediction process is calculated as follows:

$$d = x_{a2} - x_{b1} - v_{\text{e_long}} \times \frac{l_w}{v_{\text{c_lat}}} \tag{10}$$

Where x_{a2} is the transverse coordinate of the end position of the cut-in vehicle, x_{b1} is the transverse coordinate of the start position of the ego vehicle, $v_{\text{e_long}}$ is the longitudinal speed of the ego vehicle, $v_{\text{c_lat}}$ is the lateral velocity of the cut-in vehicle, and l_w is the width of each of the three lanes. The setup scenario defaults to x_{a1} being greater than x_{b1}.

D. Decision state machine model building

The traffic scenario modeled by this decision-state machine involves an emergency cut-in scenario on a three-lane high-speed road, featuring two vehicles: the ego vehicle (autonomous vehicle) positioned in the right lane and the emergency cut-in vehicle in the left lane at a certain distance ahead of the ego vehicle.

A finite-state machine, commonly referred to as a state machine, consists of three key elements: State, Event, and Action. In the context of autonomous driving, the "State" represents a finite set of predefined states corresponding to basic driving behavior patterns. The triggering "Event" is primarily determined by the driving behaviors of vehicles in adjacent lanes. The "State Transfer Function" defines the rules governing transitions between these behavior patterns and is triggered when the autonomous vehicle detects specific state-triggering events [10].

Firstly, the autonomous vehicle driving behavior mode is analyzed, according to the different motion states of the vehicle can be divided into normal driving, following, braking, lane-changing and lane-changed. The main driving functions can be completed through real-time switching between them. Combining the state transfer conditions can get the decision-making and mode-switching diagram shown in Fig. 5.

The switching between states is determined by the state transfer conditions, the state transfer conditions involved in this state machine are shown in Table 1. The specific flowchart of the algorithm is shown in Fig. 6.

III. Control Model

In this section, the lateral and longitudinal controllers are designed to receive the reference trajectory information from the decision-making module, and accurately and stably control the target volume by controlling the accelerator, brake, and steering wheel angle to complete the risk-avoidance driving operation of the autonomous vehicle in the dynamic traffic environment.

A. Lateral LQR Controller

The objective of the optimal control LQR (Linear Quadratic Regulator), based on linear quadratic, is to find the optimal control to obtain an extreme value for the following performance function:

$$J = \frac{1}{2}\int_0^{\infty} \left[e^{\mathrm{T}}(t)Qe(t) + u^{\mathrm{T}}(t)Ru(t)\right] dt \tag{11}$$

In the equation, both Q and R are weight matrices, and U represents the control input. Prior to this, we can obtain the feedback coefficients K offline by using the look-up table method to calculate the corresponding table for different vehicle speeds and K. The table can be used to obtain the feedback coefficients K by using the Hamiltonian function to calculate the corresponding table for different vehicle speeds.

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Triggering condition	Trigger condition information description
Vehicle speed	a1: The difference between the speed of the cut-in vehicle and the desired speed of the ego vehicle is less than the threshold value a2: The difference between the speed of the cut-in vehicle and the desired speed of the ego vehicle is more than the threshold value
Traffic conflict time indicator TTC	b1: TTC is less than the following threshold more than the braking threshold b2: TTC is less than the braking threshold more than the lane change threshold b3: TTC is less than lane changing threshold
Traffic conflict time indicator THW	c1: THW is more than the following threshold c2: THW is less than the following threshold more than the braking threshold c3: THW is less than the braking threshold greater than the lane change threshold c4: THW is less than the lane changing threshold
OBB collision detection information	d1: there is no collision d2: there is a collision
Longitudinal prediction distance d	e1: More than 21m e2: Less than 21m and more than 15m e3: Less than 15m and more than 5m
Ego vehicle heading angle h_c	f1: Approximately equal to 90 °
Lane position	g1: Ego vehicle in carriageway g2: Ego vehicle in the adjacent lane

TABLE I
TRIGGERING CONDITION INFORMATION

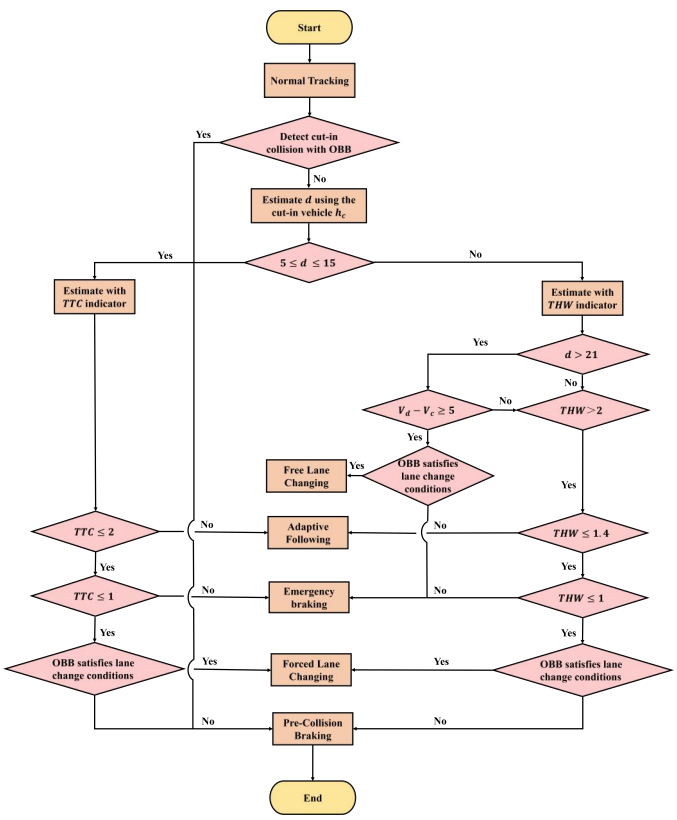

Fig. 6. Algorithm flow chart

The Hamilton function can also be used to derive the control law for LQR:

$$U(t) = -Ke(t) = -R^{-1}B^{\mathrm{T}}Pe(t) \tag{12}$$

Where P is a solution of the Riccati equation and B and e(t) can be obtained from the constructed Hamiltonian function.

Since the system suffers from a steady state error, in order to overcome its adverse effect on the system, feed forward control is introduced in this paper. The goal is to find an appropriate feed-forward control quantity with as much zero stabilisation error as possible to obtain the final full control rate:

$$U = -K_{err} + \delta_f \tag{13}$$

δ_f is the feedforward control quantity and e_{rr} is the steady state error present in the system.

By substituting the error equation to calculate and simplify, the result cannot be zero since the third row of the obtained e_{rr} matrix is not heading error, so the first row of the e_{rr} matrix can be used to design the feedforward control law:

$$\delta_f = k_r\left[a + b - bk_3 + \frac{mv_x^2}{a+b}\left(\frac{b}{c_f} + \frac{a}{c_r}\right) - k_3\left(\frac{a}{c_r}\right)\right] \tag{14}$$

Here, k_r denotes the road curvature, a represents the distance from the center of mass to the front axle, b represents the distance from the center of mass to the rear axle, k_3 is one of the gain coefficients of the LQR controller, m denotes the vehicle mass, v_x represents the longitudinal speed of the vehicle, c_r denotes the cornering stiffness of the rear wheels, and c_f denotes the cornering stiffness of the front wheels.

In addition, in order to achieve horizontal and vertical control decoupling, this paper establishes natural coordinates (Frenet Coordinate System) on the original planning trajectory. And the position information of each is shown in Fig. 7. Where point A is the true position of the vehicle, point B is the point where the true position of the vehicle is projected onto the natural coordinate system, and point C is the matching point, i.e., the nearest point found by traversing the currently planned trajectory, the matching point is equal to the projection point only when the trajectory is planned intensively, and the current position of the vehicle can thus be calculated to be the distance between the current position of the vehicle and the matching point e_s , as well as the desired speed v_{des}, as inputs to the longitudinal control. The calculation formula is specified as

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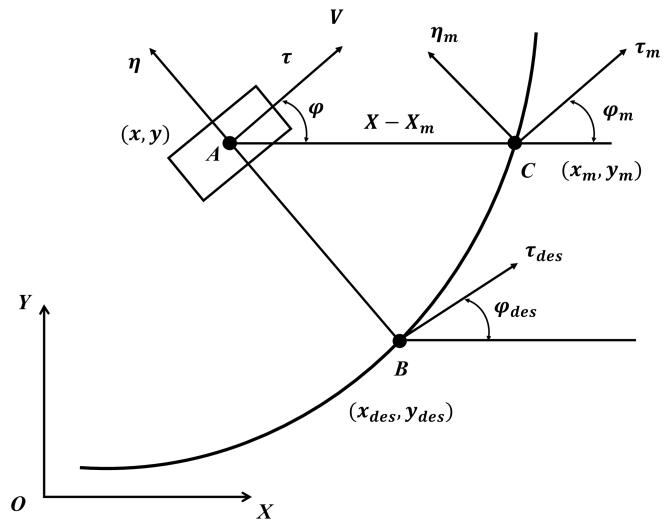

Fig. 7. Vehicle error calculation model in natural coordinate system

follows:

$$\tau_m = (\cos\varphi_m, \sin\varphi_m) \tag{15}$$

$$\eta_m = (-\sin\varphi_m, \cos\varphi_m) \tag{16}$$

$$d = X - X_m = (x - x_m, y - y_m) \tag{17}$$

$$v_{\text{des}} = \frac{v_x \cos e_\varphi - v_y \sin e_\varphi}{1 - k_r e_d} \tag{18}$$

$$e_s = d\tau_m \tag{19}$$

$$e_d = d\eta_m \tag{20}$$

$$e_v = v_r - v_{\text{des}} \tag{21}$$

In Eqs. 3.5 and 3.6, τ_m,η_m are the tangential unit vector and normal unit vector of the matching point; in Eq. 3.7, X is the real bit vector of the vehicle, X_m is the bit vector of the matching point, and d is the difference between the two vectors; and in Eq. 3.8, v_r is the actual velocity, and e_v is the difference between the desired velocity and the actual velocity. It is also possible to calculate the actual acceleration a_r [11].

B. Longitudinal PID Controller

Based on the above calculations of $e_s, v_{des}, v_r, e_v, a_r$, etc., a dual PID longitudinal control system is designed in this paper. The first PID controller is used for speed control, i.e., to reduce the speed error e_v, and the second PID is used to control the position error e_s. 'P' represents proportional, which accelerates the speed to reach the target; 'I' signifies integral, which serves to eliminate steady-state error but may cause overshooting; and 'D' denotes differential, which helps to suppress proportional overshooting but complicates the reduction of integral overshooting. Therefore, if the error is not large then the integral term is generally not applied to meet the requirement of accurate tracking.

IV. Simulation Results And Analysis

In this study, a traffic accident case with high typicality and high-risk factor collected by the National Highway Traffic Safety Administration (NHTSA) of the United States was selected [12]. Based on the real traffic accident data in the case, the initial speed of the vehicle driving as well as the initial longitudinal distance are supplemented with the description of the real accident scenario, and the relevant information and schematic diagrams about the traffic accident are shown in Table 2 and Fig. 8. Using Prescan simulation software to build an accident scenario that occurred in 2018 where an adjacent cut-in vehicle collided with the rear of a high-speed straight vehicle, and a decision-state machine model was constructed in Simulink to control the behavior of the straight vehicle, and then a joint simulation study was carried out by using Prescan&Matlab/Simulink&Carsim, and through the comparative experiments, the effectiveness and performance advantages of the proposed autonomous vehicle decision-making approach are verified under the emergency cut-in scenario.

TABLE II
Case Information

Type of Case Information	Case Information
Case Number	CR 18015
Crash Date	4/2018
Crash Site	New York
Vehicle 1 Model	2014 Audi Q5
Vehicle 2 Model	2006 Acura MDX
Posted Speed Limit	72 kmph
Vehicle 1 Longitudinal Velocity Delta	12 kmph
Vehicle 1 Lateral Velocity Delta	0 kmph
Vehicle 1 Barrier Equivalent Speed	7 kmph
Vehicle 2 Longitudinal Velocity Delta	-13 kmph
Vehicle 2 Lateral Velocity Delta	0 kmph
Vehicle 2 Barrier Equivalent Speed	16 kmph
Vehicle 1 Initial Velocity	10 m/s
Vehicle 2 Initial Velocity	25 m/s
Longitudinal Distance	10 m

In the simulation experiment, the scenario is set on a three-lane motorway where the initial longitudinal distance between the emergency cut-in vehicle and the ego vehicle is 10 meters. The cut-in vehicle decelerates at a rate of 10 m/s², while its longitudinal speed increases by 3.33 m/s and the transverse speed remains constant. Meanwhile, the ego vehicle starts with an initial speed of 25 m/s, with its longitudinal speed decreasing by 3.61 m/s and its transverse speed also remaining unchanged.

A. Decision State Machine Model Considering Only Adaptive Following

The state machine model consists of three sub-states: Acceleration, Following and Deceleration. Under the decision-state machine model, the vehicle (starred vehicle) first takes deceleration measures to increase the headway distance after the front vehicle cuts in, and then follows the vehicle, but it fails to avoid a collision with the vehicle in front of it. This is because the longitudinal distance between the cut-in vehicle and the ego vehicle is too close after the cut-in, and at the same time, the ego vehicle travels at a higher speed, so purely decelerating and braking can't avoid the risk in time.

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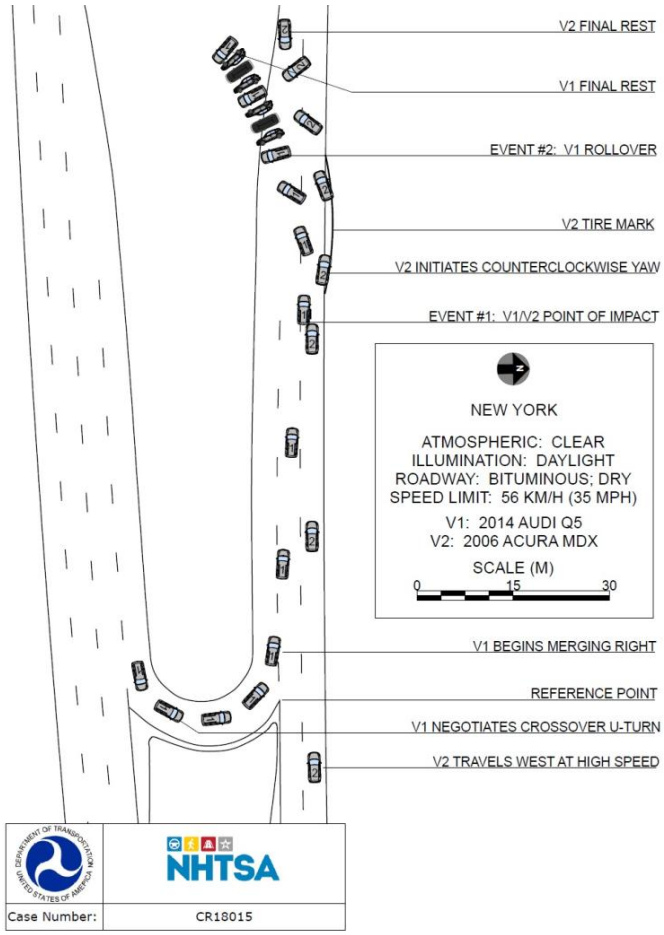

Fig. 8. Schematic diagram of a traffic accident scene

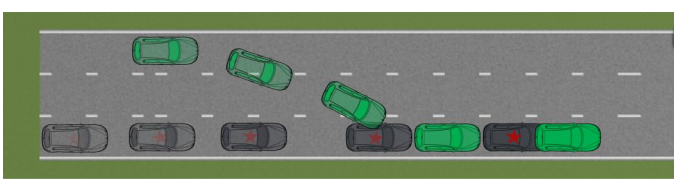

Fig. 9. Schematic diagram of the decision simulation process considering only the adaptive following state machine

The driving behavior of the autonomous vehicle after decision-making is shown in Fig. 9.

B. Decision State Machine Model Designed in This Paper

The decision state machine model in this paper consists of three substates: NormalTracking, Lanekeep, and LaneChange. LaneChange is further divided into LaneChanging and LaneChanged, while Lanekeep is divided into Following and Braking. The state machine first considers the transient process of the cut-in scenario to predict the behavior of the cut-in vehicle. In the scenario shown in Figure 10, the longitudinal distance between the two vehicles after the cut-in is predicted to be less than the threshold distance according to the longitudinal prediction distance formula in Section 2 [13].

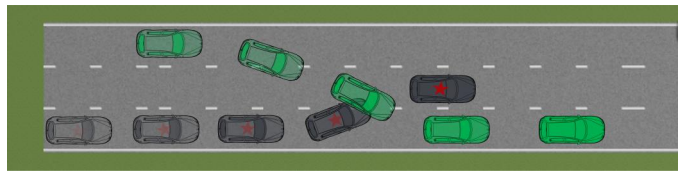

Fig. 10. Schematic diagram of the decision simulation process for designing state machines in this paper

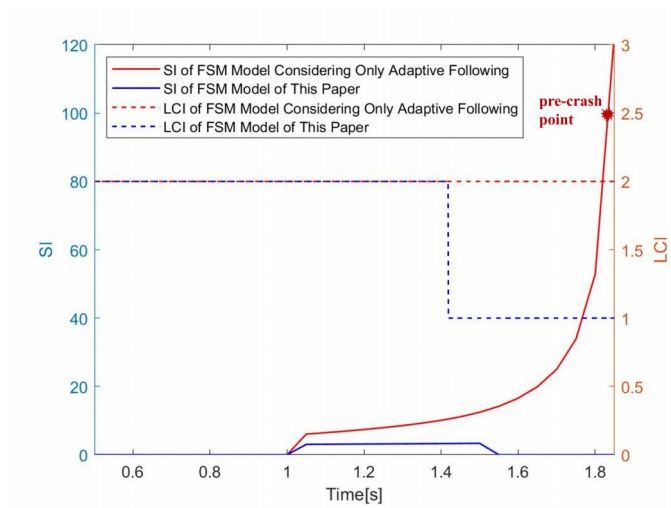

Fig. 11. Schematic comparison of the results of the two decision state machines

This indicates that simply decelerating and braking cannot avoid the risk of collision. Subsequently, the Oriented Bracket Box (OBB) collision detection method is used to determine whether lane changing will lead to a collision with a potential left rear vehicle, i.e., whether the autonomous vehicle meets the conditions for lane changing. Therefore, the decision state machine adopts mandatory lane changing measures to help the vehicle avoid collisions and ensure the safety of drivers and passengers.

Calculate the longitudinal safety risk index value SI and lane change information value for the ego and cut-in vehicles under the two decision-state machine models. The lane change information value, referred to as LCI, is 2 for lane keeping and 1 for lane changing. Obtain the comparison graphs of SI and LCI shown in Fig. 11.

As can be seen from the trend graphs of safety indicator and lane information, under the decision-state machine model that only considers the adaptive following, the ego vehicle fails to avoid a collision with the cut-in vehicle, whereas under the decision statedecision-state machine model in this paper, the ego vehicle avoids a collision with the cut-in vehicle when it switches to the adjacent lane and continues to drive at about 1.4s. In addition, this study also derives the driving behavior of the ego vehicle after the two state machine decisions in the case of different initial longitudinal distances between the cut-in vehicle and the ego vehicle, as shown in Figures 12 and 13.

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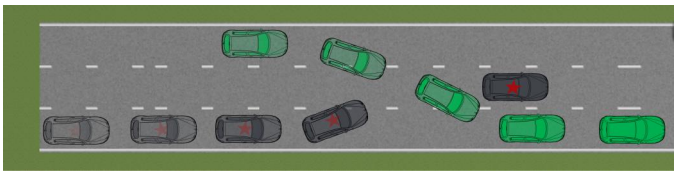

Fig. 12. Schematic diagram of the decision-making simulation of free lane-changing behavior

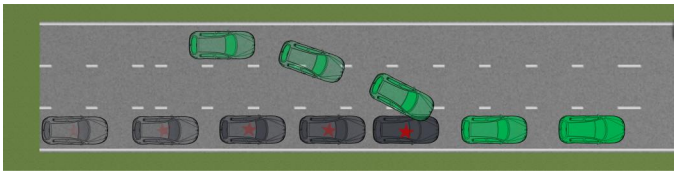

Fig. 13. Schematic diagram of adaptive following behavior decision simulation

Figure 12 illustrates that when the longitudinal distance is 22 meters, the ego vehicle switches to the adjacent lane to accelerate further in pursuit of a higher speed. Whereas, Figure 13 shows that when the longitudinal distance is 15 meters, the ego vehicle adopts adaptive following, slowing down initially and then adjusting its speed to maintain a safe distance.

From the simulation results of the two scenarios mentioned above, it can be concluded that the decision-state machine model designed in this paper can help the vehicle to complete the collision avoidance, while also considering the driving efficiency.

V. Conclusions

This paper presents a safety risk evaluation based autonomous vehicle decision-making approach for cut-in emergency scenario. The approach addresses the limitations of traffic conflict time indicators such as TTC and THW. It uses an Oriented Bounding Box (OBB) collision detection approach and a longitudinal distance prediction formula. This approach predicts the cut-in behavior based on the transient process of the scenario. It dynamically adjusts the decision-making behavior of the autonomous vehicle by incorporating these predictions as key parameters in the decision model. The decision state machine model was constructed using real traffic accident data from NHTSA. This enhances the reliability and applicability of the approach. Joint simulations were conducted in Prescan, Matlab/Simulink, and Carsim. These simulations compared the behavior decision results of different decision state machines. The results show that the proposed approach can effectively assess and mitigate collision risks in emergency cut-in scenarios. Future work will focus on more complex cut-in scenarios and developing more efficient decision algorithms by integrating learning algorithms.

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